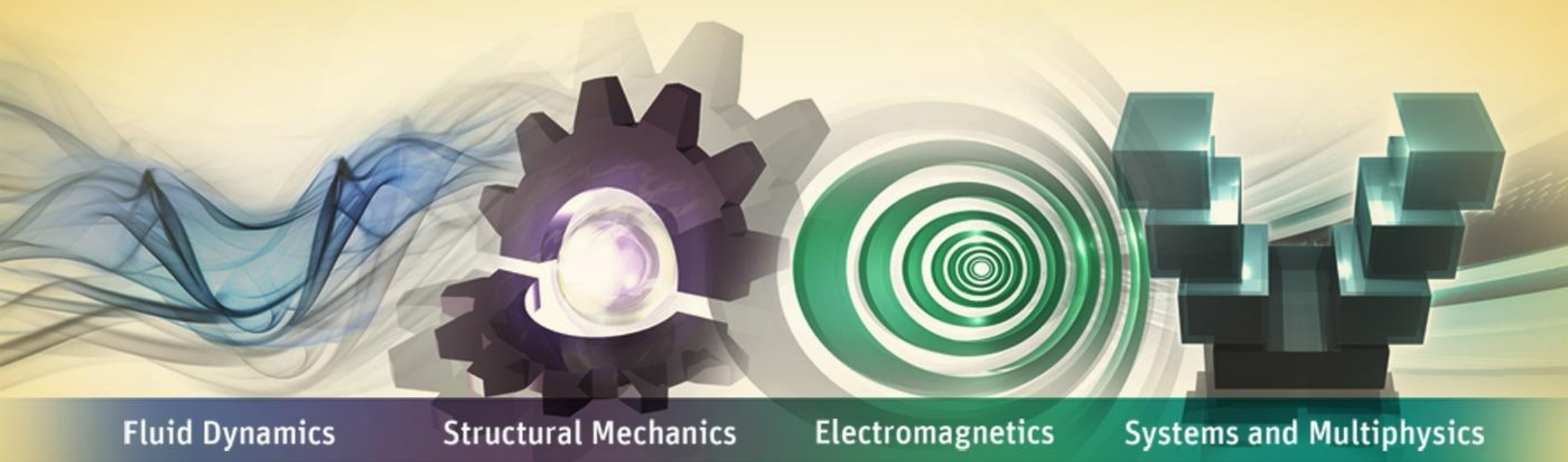


Lecture 3: Contacts



ANSYS Mechanical

Contact:

- When two separate surfaces touch each other such that they become mutually tangent, they are said to be in *contact*.
- In the common physical sense, surfaces that are in contact have these characteristics:
 - They do not interpenetrate.
 - They can transmit compressive normal forces and tangential friction forces.
 - They often do not transmit tensile normal forces.
 - Surfaces are free to separate and move away from each other.
- Contact is a *changing-status* nonlinearity. That is, the stiffness of the system depends on the contact status, whether parts are touching or separated.

Basic Types of contact behaviors available*:

Bonded: No separation and no sliding between faces or edges.

No Separation: Similar to bonded, except frictionless sliding can occur along contacting faces.

Frictionless: Surfaces are free to slide and separate without resistance.

Rough: Similar to the frictionless setting except no sliding allowed.

Frictional: Allows sliding with resistance proportional to user defined coefficient of friction, with freedom to separate without resistance.

The implementation of these behaviors will be presented in detail in following slides of this lecture and the next.

* More advanced behaviors are available thru MAPDL commands. Refer to Chapter 3 of Advanced Connections Course for details.

NORMAL DIRECTION BEHAVIOUR



FRICTIONLESS CONTACT

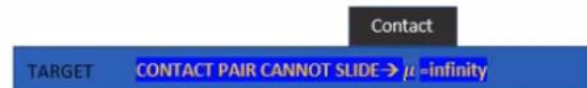
FRICTIONAL CONTACT

ROUGH CONTACT

NO-SEPERATION

BONDED

TANGENTIAL DIRECTION BEHAVIOUR



Contact Means No Penetration



Non Linear Contact because it can separate in normal direction

Details of "Bonded - SYS\Solid To SYS\Solid"

Scope	
Scoping Method	Geometry Selection
Contact	1 Face
Target	1 Face
Contact Bodies	SYS\Solid
Target Bodies	SYS\Solid
Definition	
Type	Bonded
Scope Mode	Bonded
Behavior	No Separation
Trim Contact	Frictionless
Suppressed	Rough
	Frictional

Linear Contact because it cannot separate in normal direction

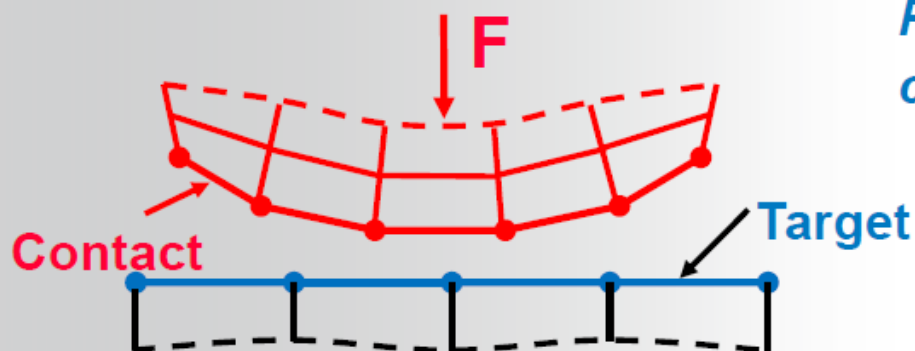
Table 1: Contact Names and Behavior

Name	Gap Open/Close?	Sliding Allowed?
Bonded	No	No
Rough	Yes	No, infinite μ
No Separation	No	Yes, $\mu = 0$
Frictionless	Yes	Yes, $\mu = 0$
Frictional	Yes	Yes, if $F_{\text{sliding}} > F_{\text{friction}}$

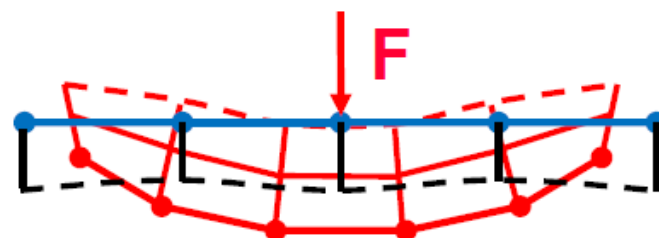
How compatibility is enforced in a contact region:

- Physical contacting bodies do not interpenetrate. Therefore, the program must establish a relationship between the two surfaces to prevent them from passing through each other in the analysis.
- When the program prevents interpenetration, we say that it *enforces contact compatibility*.
- Mechanical* offers several different *contact formulations* to enforce compatibility at the contact interface.

Details of "Frictionless - Solid To Solid"	
[-] Scope	
Scoping Method	Geometry Selection
Contact	1 Face
Target	1 Face
Contact Bodies	Solid
Target Bodies	Solid
[-] Definition	
Type	Frictionless
Scope Mode	Automatic
Behavior	Program Controlled
Suppressed	No
[-] Advanced	
Formulation	Program Controlled
Detection Method	Program Controlled
Interface Treatment	Augmented Lagrange
☐ Offset	Pure Penalty
Normal Stiffness	MPC
Update Stiffness	Normal Lagrange
	Mouse



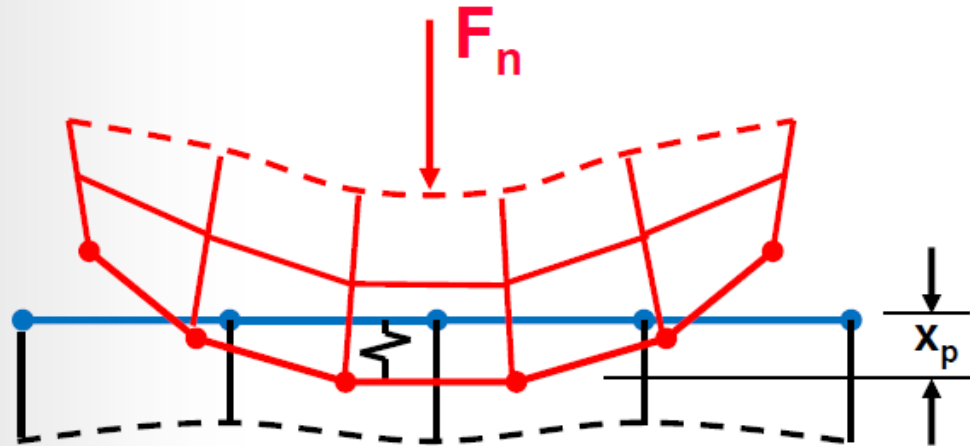
Penetration occurs when contact compatibility is not enforced.



B. Contact Formulations

- *Pure Penalty:*

$$F_{normal} = k_{normal} x_{penetration}$$



- Here, for a finite contact force F_{normal} , there is a concept of contact stiffness k_{normal} . The higher the contact stiffness, the lower the penetration $x_{penetration}$, as shown in the figure.
- Ideally, for an infinite k_{normal} , one would get zero penetration. This is not numerically possible with penalty-based methods, but as long as $x_{penetration}$ is small or negligible, the solution results will be accurate.

How to chose the right k ?

Too low of a value of k_n results in large penetration x_p

One can imagine that if the penetration is on the same order of magnitude as the calculated displacements, the results will be quite suspect.

Selection of k_n that is too high leads to convergence difficulties

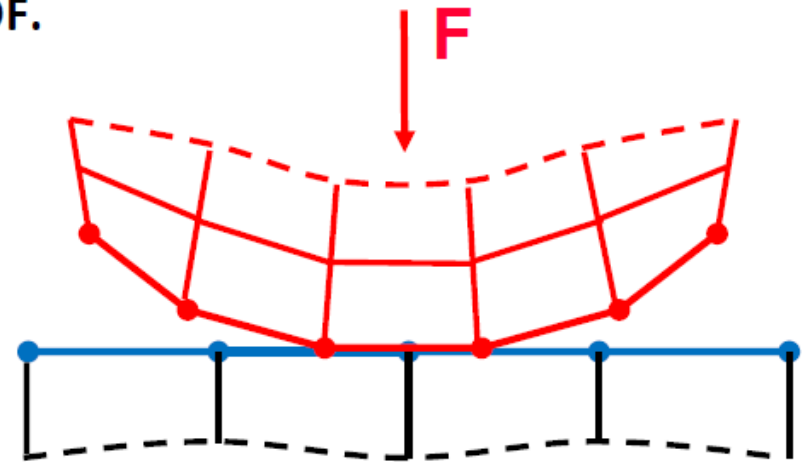
The good news is that, by default, ANSYS automatically calculates contact stiffness k_n , based on the underlying solid element's size and material properties. The analyst is left with the task of monitoring the nonlinear solution — if convergence difficulties are met due to too high of an k_n value, ANSYS provides tools on determining which contact region's contact stiffness should be reduced. Likewise, postprocessing penetration x_p is available after a converged solution to ensure that k_n was not too low.

For experienced users wishing to manually change the default contact stiffness, in the Details view of a Contact Region, the “Advanced: Normal Stiffness” option should be changed to “Manual”, and the user may input a “Normal Stiffness Factor”.

- *Normal Lagrange:*

- Adds an extra degree of freedom (contact pressure) to satisfy contact compatibility. Consequently, instead of resolving contact force as contact stiffness and penetration, contact force (contact pressure) is solved for explicitly as an extra DOF.

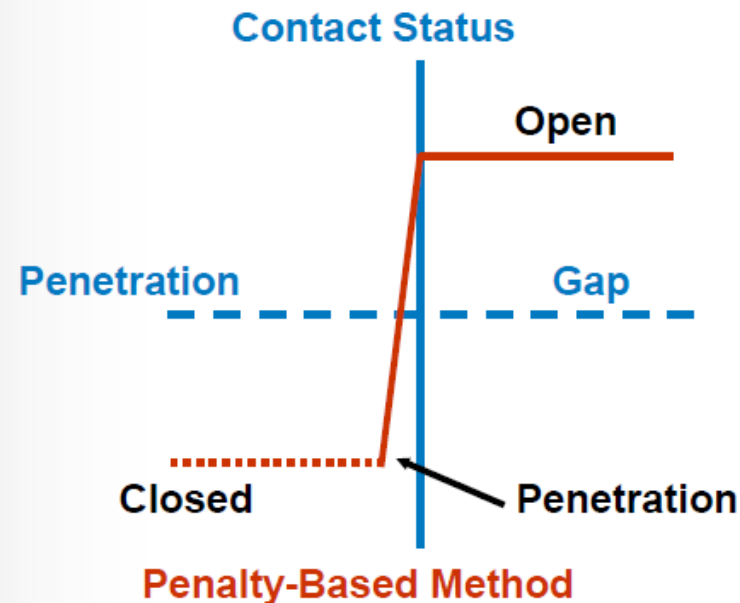
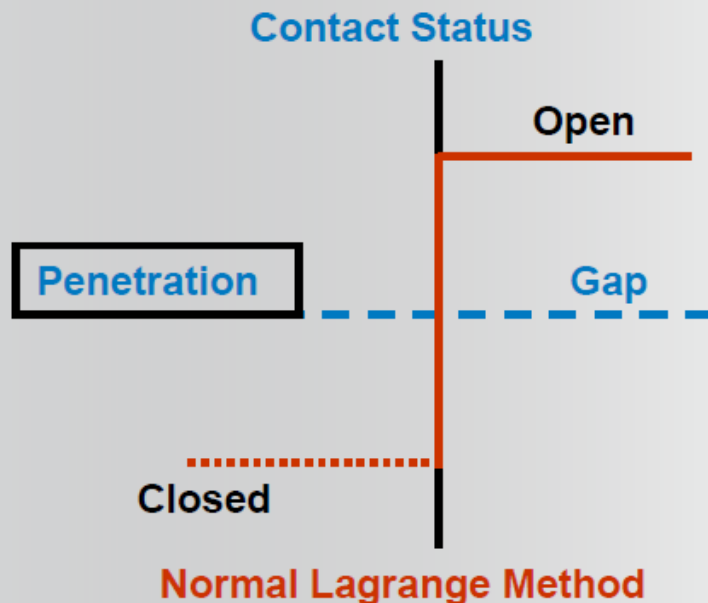
$$F_{normal} = DOF$$



- Enforces zero/nearly-zero penetration with pressure DOF
- Does *not* require a normal contact stiffness (zero elastic slip)
- Requires Direct Solver, which can be more computationally expensive
- Only applies to forces in directions Normal to contact surface

Chattering is an issue which often occurs with Normal Lagrange method

- If no penetration is allowed (left), then the contact status is either open or closed (a step function). This can sometimes make convergence more difficult because contact points may oscillate between open/closed status. This is called *chattering*
- If some slight penetration is allowed (right), it can make it easier to converge since contact is no longer a step change.



- *Augmented Lagrange*: A penalty based method that “augments” the pure penalty calculation with a extra Lagrangian term:

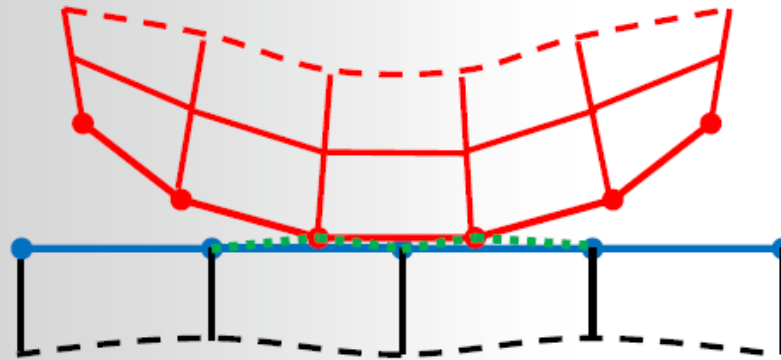
Augmented Lagrange: $F_{normal} = k_{normal} x_{penetration} + \lambda$



- Because of the extra term λ , the augmented Lagrange method is less sensitive to the magnitude of the contact stiffness k_{normal} .
- Augmented Lagrange is the default formulation used for *Program Controlled* option

- *Multi-Point Constraint (MPC) Formulation:*

- Internally adds constraint equations to “tie” the displacements between contacting surfaces



- This approach is not penalty-based or Lagrange multiplier-based. It is a direct, efficient way of relating surfaces of contact regions which are bonded.
- Large-deformation effects also are supported with MPC-based bonded contact
- Applies specifically to “Bonded” and “No Separation” Types of contact

- **Augmented Lagrange** (Default): Suitable for most problems.
- **Pure Penalty**: Contact occurring only on Edge or Corner
- **MPC** (Multi-Point Constraint): Ideal for all bonded/no-separation contact when there is no over-constraint
- **Beam**: this option utilizes internally-created massless linear beam elements to connect the two sides of a contact interface together. This can be more efficient than the traditional formulations and can avoid the over constraints that can happen if multiple contact regions utilizing the MPC option end up generating constraint equations that tend to conflict with each other.
- **Normal Lagrange**:
 - Highest accuracy
 - Contact with material nonlinearities
 - Between shells or thin layers
 - Interference fit
 - Large Sliding

Table 2: Formulation Options

Name	Equations Solved	Default Contact Detection
MPC	Generates CE's*	Nodes
Pure Penalty	$F = Kx$	Gauss
Augmented Lagrange	$F = Kx + \lambda$	Gauss
Normal Lagrange	Nodal Pressure DOF	Node



*CE: Constraint Equations

Nodes

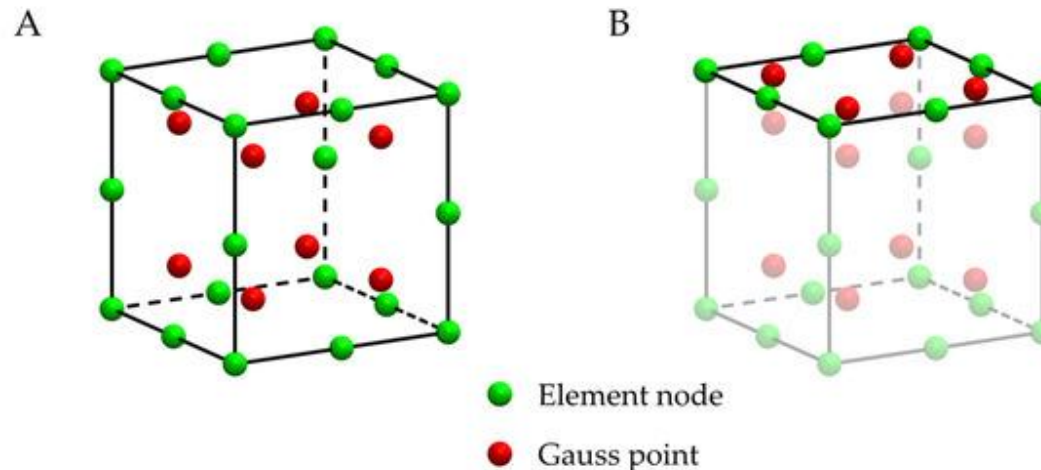
Corners or edges of an element. They are the primary locations where the solver calculates **primary unknowns**, such as:

- **Displacements** (in structural analysis)
- **Temperatures** (in thermal analysis)
- **Pressures** (in fluid analysis)

Gauss Points

Mathematical locations *inside* the element where the actual integration of the governing equations happens. ANSYS uses "Gaussian Quadrature" to calculate:

- **Stiffness matrices**
- **Strains and Stresses** (derived from nodal displacements)
- **Heat flux**



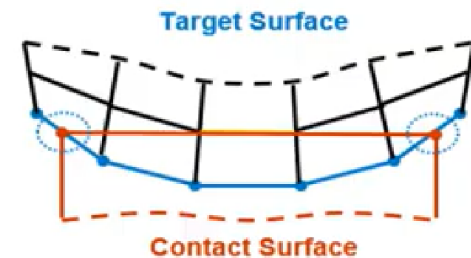
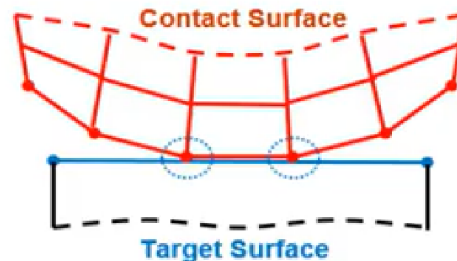
Most commonly used Non-linear Contacts Settings

1- Scoping (Contact VS Target)

Details of "Frictional - SYS\Solid To SYS\Solid"	
Scope	
Scoping Method	Geometry Selection
Contact	1 Face
Target	1 Face
Contact Bodies	SYS\Solid
Target Bodies	SYS\Solid
Protected	No
Definition	
Type	Frictional
☐ Friction Coefficient	0.15
Scope Mode	Manual
Behavior	Asymmetric
Trim Contact	Off
Suppressed	No
Advanced	
Formulation	Pure Penalty
Small Sliding	Program Controlled
Detection Method	Program Controlled
Penetration Tolerance	Program Controlled
Elastic Slip Tolerance	Program Controlled
Normal Stiffness	Factor
Normal Stiffness Factor	0.1
Update Stiffness	Each Iteration
Stabilization Damping Factor	0.
Pinball Region	Program Controlled
Time Step Controls	None
Geometric Modification	
Interface Treatment	Adjust to Touch
Contact Geometry Correction	None
Target Geometry Correction	None

If the the two surfaces are different in:

- **Shape:** If a convex surface comes into contact with a flat or concave surface, the **flat or concave** surface should be the **Target**.
- **Mesh size:** The surface with the **coarser** mesh should be the **Target**.
- **Stiffness:** The **stiffer** surface should be the **Target**.
- **Element order:** The **lower order** surface should be the **Target**.
- **Geometrical size:** The **larger** surface should be the **Target**.

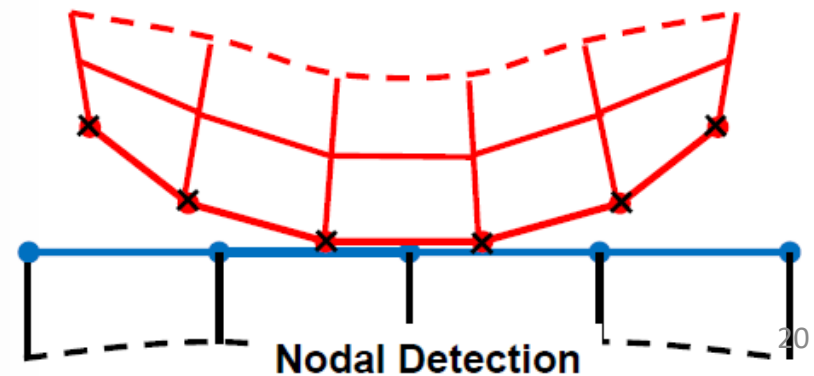
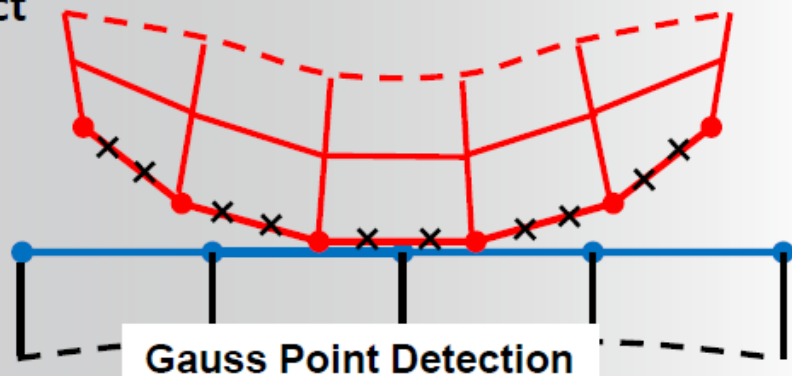


C. Detection Method

Detection Method allows you to choose the location of contact detection used in the analysis in order to obtain a good convergence.

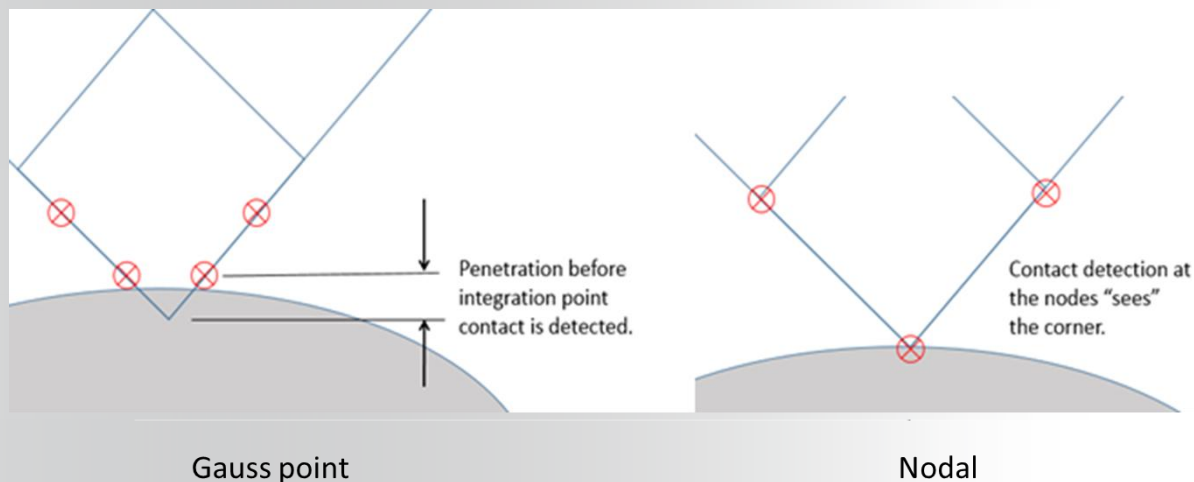
- Pure Penalty and Augmented Lagrange Formulations use *Gauss point detection* by default. This results in more detection points (10 in this example on left) and is generally considered more accurate than nodal detection.
- Normal Lagrange and MPC Formulation use *Nodal- Normal to Target* by default. This results in fewer detection points (6 in the example on right)
- Options are applicable to 3D face-face and 2D edge-edge contact

Definition	
Type	Frictionless
Scope Mode	Automatic
Behavior	Program Controlled
Trim Contact	Program Controlled
Trim Tolerance	6.0296e-003 in
Suppressed	No
Advanced	
Formulation	Program Controlled
Detection Method	Program Controlled
Penetration Tolerance	Program Controlled
Interface Treatment	On Gauss Point
☐ Offset	Nodal-Normal From Contact
Normal Stiffness	Nodal-Normal To Target
Update Stiffness	Nodal-Projected Normal From Contact
Stabilization Damping Factor	Program Controlled
Pinball Region	Program Controlled
Time Step Control	None



It is sometimes necessary to force a *Nodal* detection method:

- Should only be used for corner or edge contact



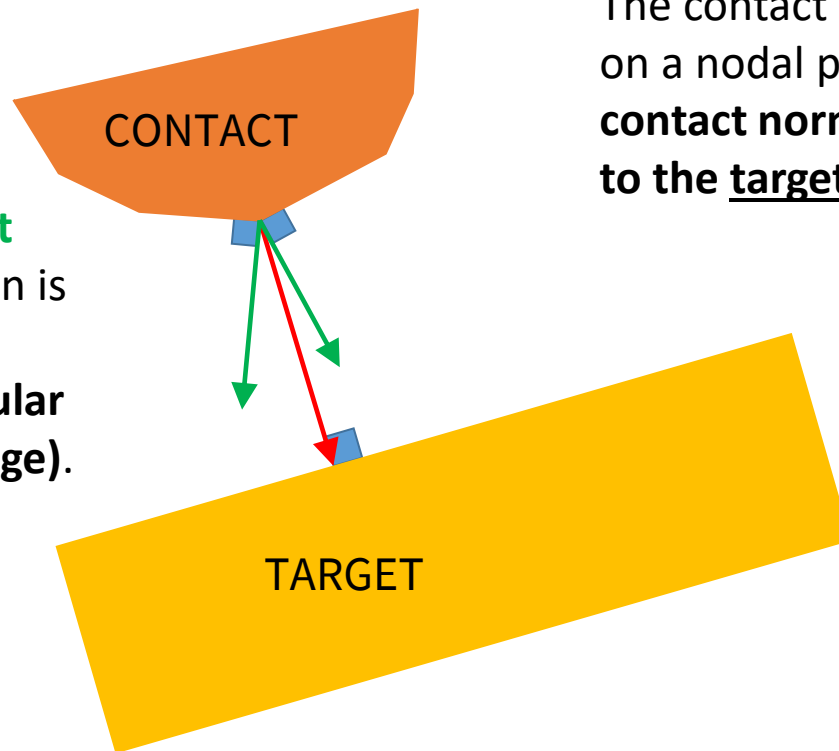
Definition	
Type	Frictionless
Scope Mode	Automatic
Behavior	Program Controlled
Trim Contact	Program Controlled
Trim Tolerance	6.0296e-003 in
Suppressed	No
Advanced	
Formulation	Program Controlled
Detection Method	Program Controlled
Penetration Tolerance	Program Controlled
Interface Treatment	On Gauss Point
☐ Offset	Nodal-Normal From Contact Nodal-Normal To Target Nodal-Projected Normal From Contact
Normal Stiffness	
Update Stiffness	Program Controlled
Stabilization Damping Factor	0.
Pinball Region	Program Controlled
Time Step Controls	None

- Normal from Contact* or *Normal to Target* dictates the direction of forces to be applied at the interface. This usually requires extra calculations to determine correct "Normal" direction. Hence, gauss detection is preferred whenever possible.

When two bodies make contact, the solver must create an equal and opposite force to prevent penetration, but what direction should that force be applied? Consider a 2D case of a flat face on one side of the contact pair and two facets on the other side of the contact pair. A single node on one element is touching the face of the element on the other side.

Nodal – Normal from contact

The contact detection location is on a nodal point where the **contact normal is perpendicular to the contact surface (average)**.



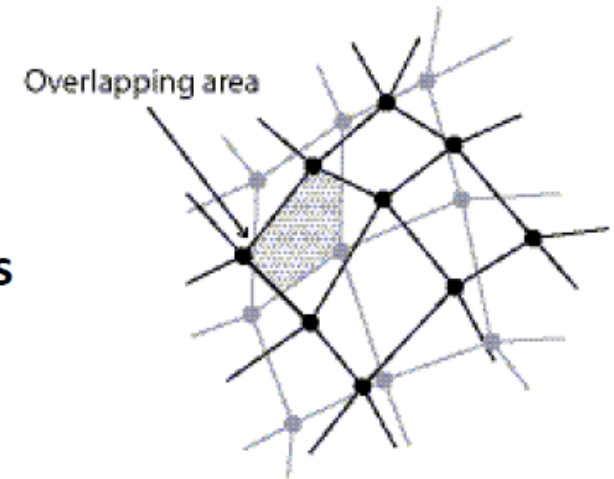
Nodal – Normal to target

The contact detection location is on a nodal point where the **contact normal is perpendicular to the target surface**.

Nodal-Projection Normal from Contact:

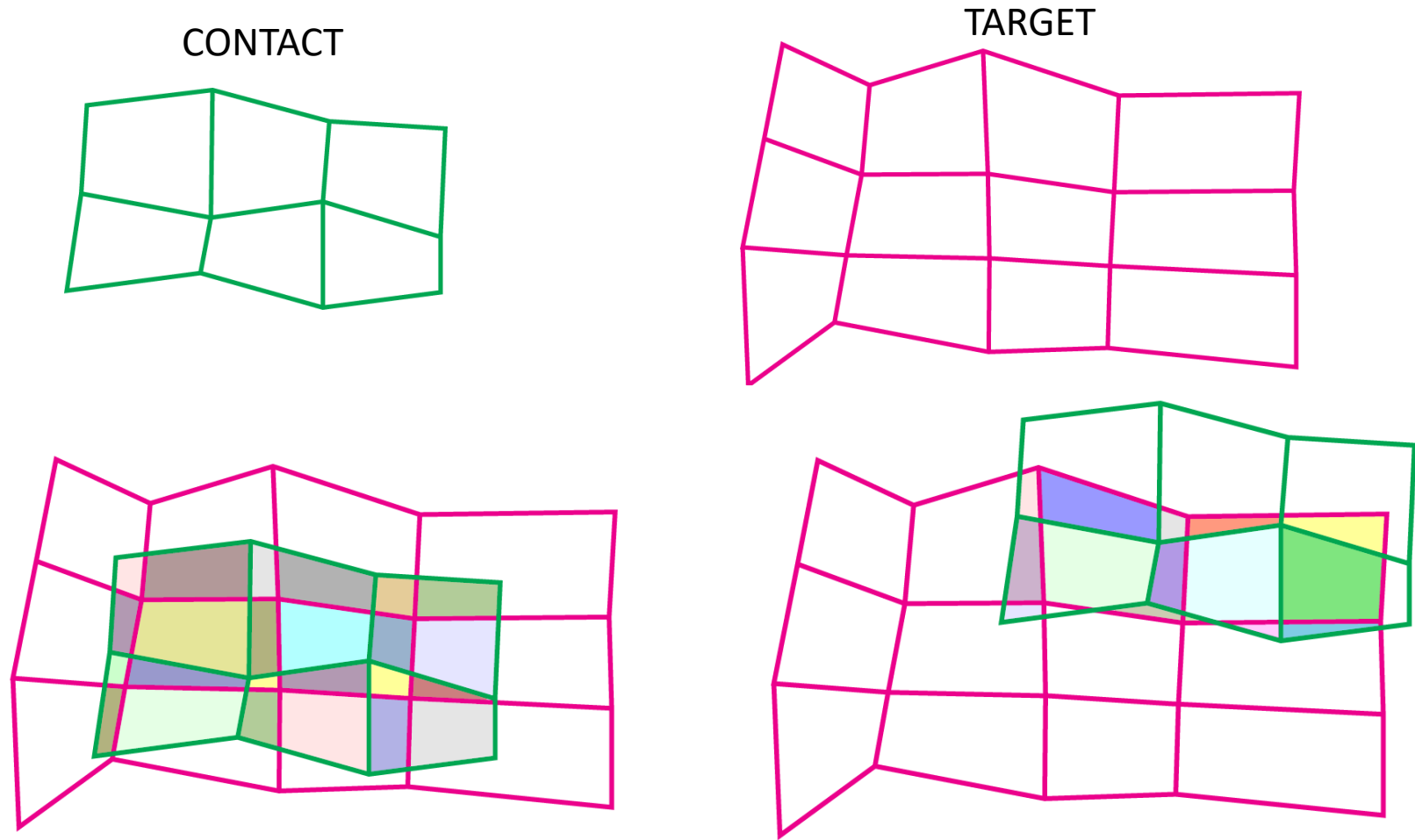
- Enforces a contact constraint on an overlapping region of the contact and target surfaces. The contact penetration/gap is computed over the overlapping region in an average sense.
- It provides more accurate contact tractions and stresses of underlying elements compared with other settings.
- Results are less sensitive to the designation of the contact and target surface.
- It satisfies moment equilibrium when an offset exists between contact and target surfaces with friction.
- Calculated Contact force distribution has smoother variation across multiple target elements.

Definition	
Type	Frictionless
Scope Mode	Automatic
Behavior	Program Controlled
Trim Contact	Program Controlled
Trim Tolerance	6.0296E-003 in
Suppressed	No
Advanced	
Formulation	Program Controlled
Detection Method	Program Controlled
Penetration Tolerance	Program Controlled
Interface Treatment	On Gauss Point
☐ Offset	Nodal-Normal From Contact
Normal Stiffness	Nodal-Projected Normal From Contact
Update Stiffness	Program Controlled
Stabilization Damping Factor	0
Pinball Region	Program Controlled
Time Step Controls	None



Nodal – Projection normal from contact

The contact detection location is at contact nodal points in an overlapping region of the contact and target surfaces (projection-based method).



Instead of calculating penetration/gap at each integration point or node, it calculates an average value for each overlapping area

Property	Description
Program Controlled	This is the default setting. The application uses Gauss integration points (On Gauss Point) when the formulation is set to Pure Penalty and Augmented Lagrange. It uses nodal point (Nodal-Normal to Target) for MPC and Normal Lagrange formulations.
On Gauss Point	The contact detection location is at the Gauss integration points. This option is not applicable to contacts with MPC or Normal Lagrange formulation.
Nodal - Normal From Contact	The contact detection location is on a nodal point where the contact normal is perpendicular to the contact surface.
Nodal - Normal To Target	The contact detection location is on a nodal point where the contact normal is perpendicular to the target surface.
Nodal - Projected Normal From Contact	The contact detection location is at contact nodal points in an overlapping region of the contact and target surfaces (projection-based method).

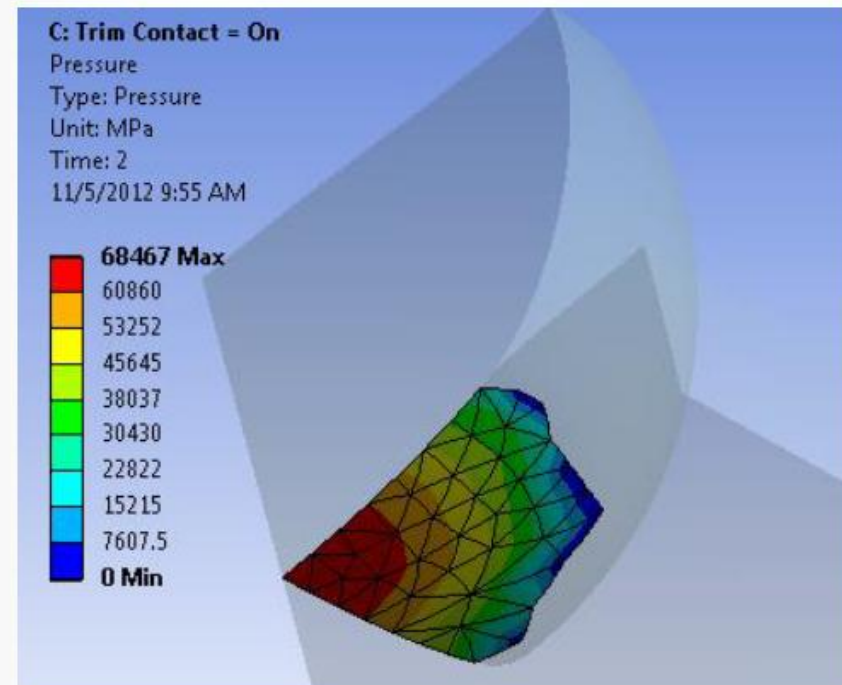
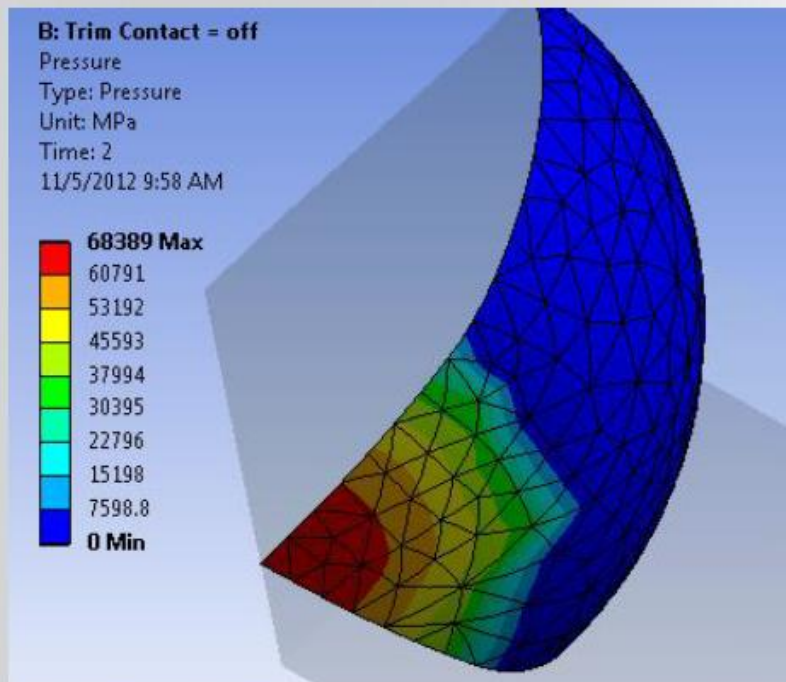
D. Trim Contact

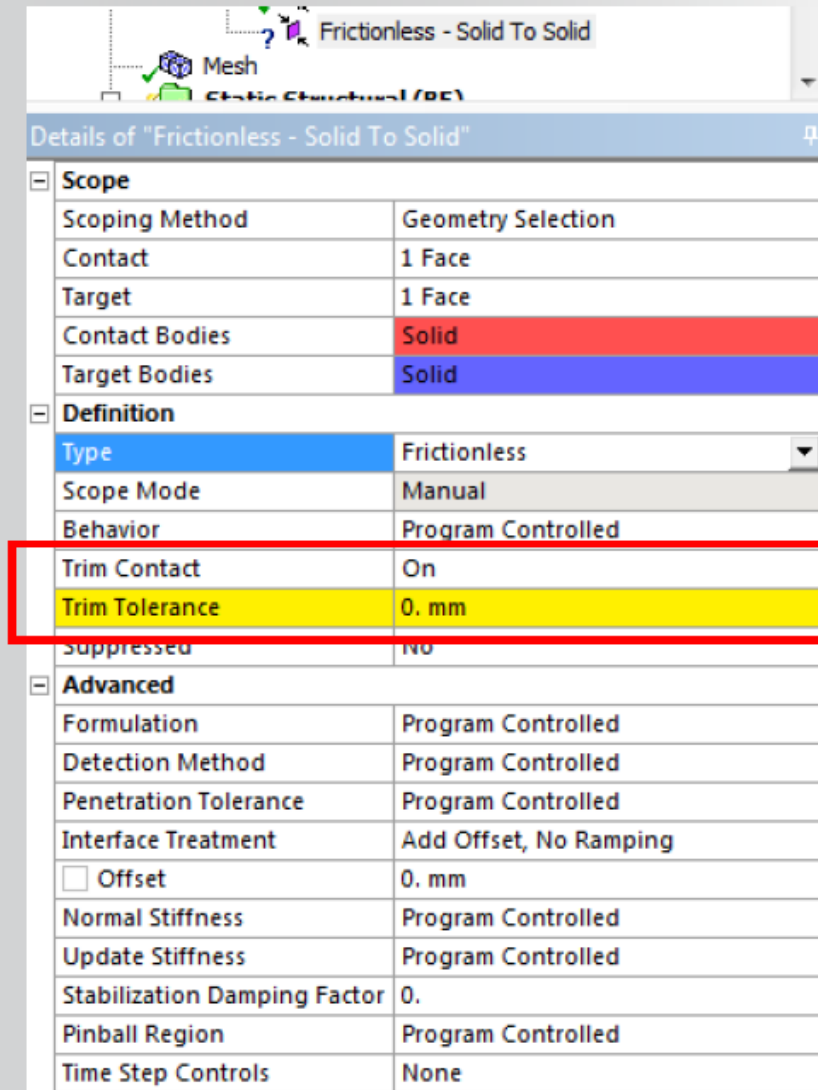
“Trim Contact” automatically reduces the number of contact elements generated within each pair, thereby speeding up processor time.

“Program Controlled” will typically turn Trim Contact ON. However, no trimming is done for manually created pairs.

Definition	
Type	Frictionless
Scope Mode	Automatic
Behavior	Program Controlled
Trim Contact	Program Controlled
Trim Tolerance	6.0296e-003 in
Suppressed	No

Consider turning Trim Contact off for large deflection sliding



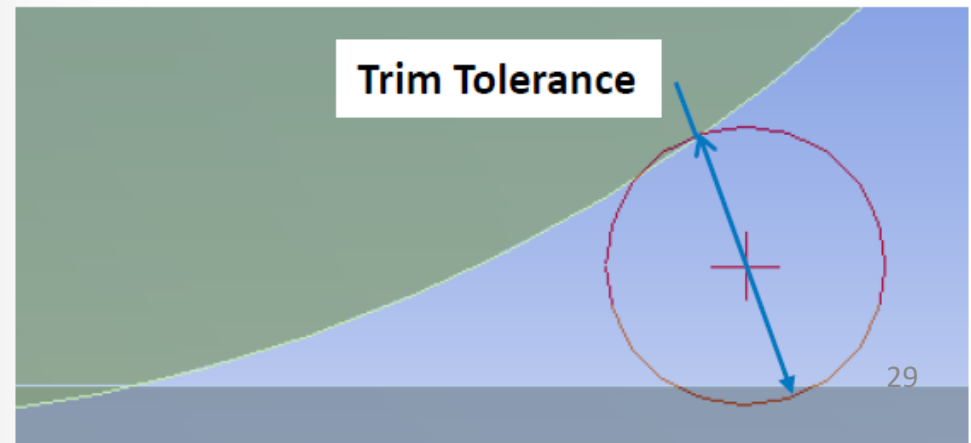


Trim Tolerance:

Defines the upper bounding box dimension used for the trimming operation.

For automatic contacts, this property displays the value that was used for contact detection and it is a read-only field.

For manual contacts, user can enter any value greater than zero.

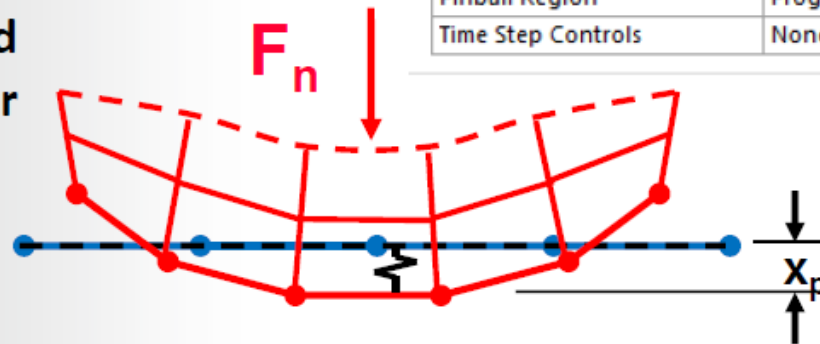


E. Penetration Tolerances

Penetration Tolerance:

- Contact compatibility is satisfied in normal direction if normal penetration (X_p) is within allowable tolerance (TOLN)
 - Can be defined as a *Factor* (of underlying element depth) or as a *Value*.
 - Default = $0.1 \times \text{element depth (surf-surf)}$
- Only applies to Penalty based formulations (Pure Penalty or Augmented Lagrange)

Advanced	
Formulation	Program Controlled
Detection Method	Program Controlled
Penetration Tolerance	Program Controlled
Elastic Slip Tolerance	Program Controlled
Interface Treatment	Add Offset, No Ramping
☐ Offset	0. mm
Normal Stiffness	Program Controlled
Update Stiffness	Program Controlled
Stabilization Damping Factor	0.
Pinball Region	Program Controlled
Time Step Controls	None



Pure Penalty:

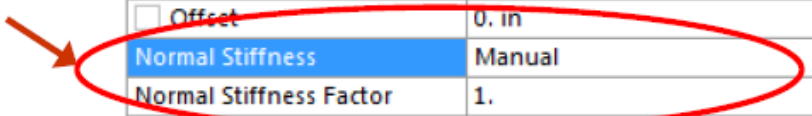
$$F_{normal} = k_{normal} x_{penetration}$$

Augmented Lagrange:

$$F_{normal} = k_{normal} x_{penetration} + \lambda$$

F. Contact Stiffness

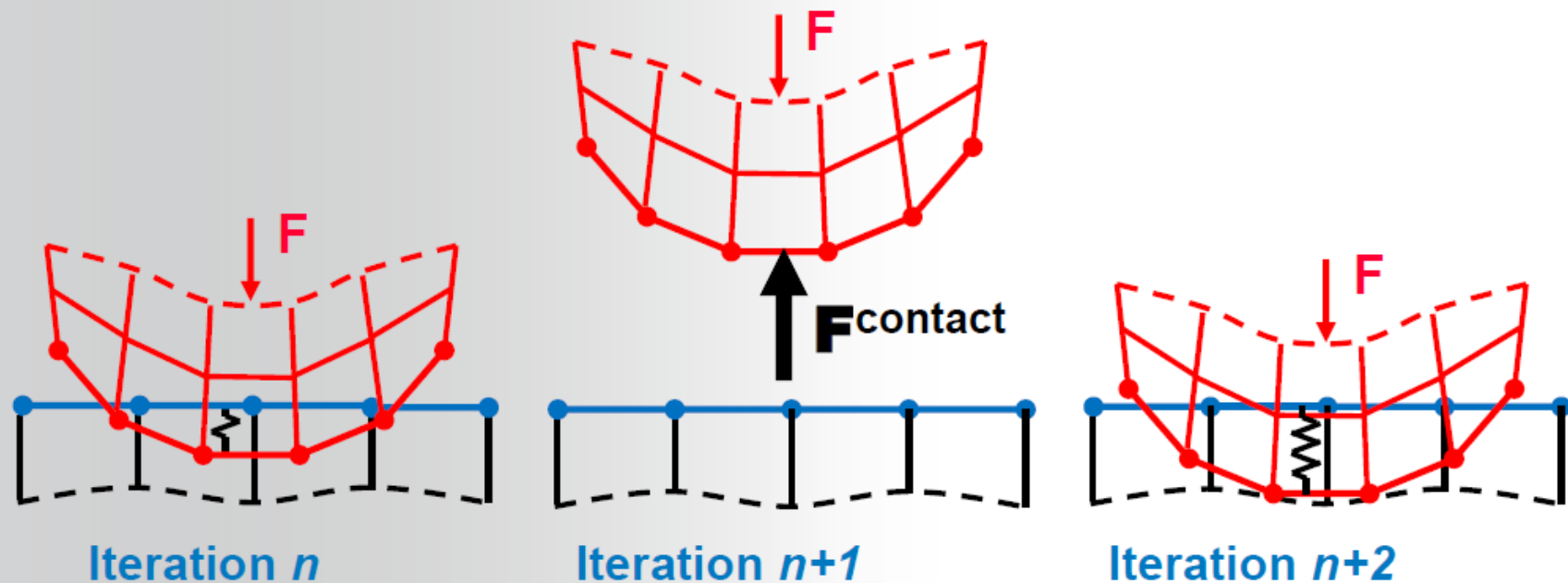
- *Normal Stiffness* is actually a multiplier or factor (FKN) on the code calculated stiffness explained earlier.
 - FKN=10 by default for bonded and no-separation behaviors
 - FKN=1.0 by default for all other behaviors
 - For bending-dominated situations, if convergence difficulties are encountered, a smaller value (FKN =0.01 - 0.1) may be helpful.
 - Only applies to Penalty based formulations (Pure Penalty or Augmented Lagrange)



Advanced	
Formulation	Program Controlled
Detection Method	Program Controlled
Penetration Tolerance	Program Controlled
Interface Treatment	Add Offset, No Ramping
☐ Offset	0. in
Normal Stiffness	Manual
Normal Stiffness Factor	1.
Update Stiffness	Program Controlled
Stabilization Damping Factor	0.
Pinball Region	Program Controlled
Time Step Controls	None

... Contact Stiffness

- The *Normal Contact Stiffness* k_{normal} is the most important parameter affecting both accuracy and convergence behavior.
 - A large value of stiffness gives better accuracy, but the problem may become more difficult to convergence.
 - If the contact stiffness is too large, the model may oscillate, with contacting surfaces bouncing off of each other



The normal stiffness can also be automatically adjusted during the solution to enhance convergence. If difficulties arise, the stiffness will be reduced automatically.

- By default, *Update Stiffness* will occur at the end of each equilibrium iteration.
- The “*Each Iteration, Aggressive*” option allows for a broader range of adjustment.



[-] Definition	
Type	Bonded
Scope Mode	Manual
Behavior	Program Controlled
Suppressed	No
[-] Advanced	
Formulation	Program Controlled
Detection Method	Program Controlled
Normal Stiffness	Program Controlled
Update Stiffness	Program Controlled
Pinball Region	Program Controlled

Never

Each Iteration

Each Iteration, Aggressive

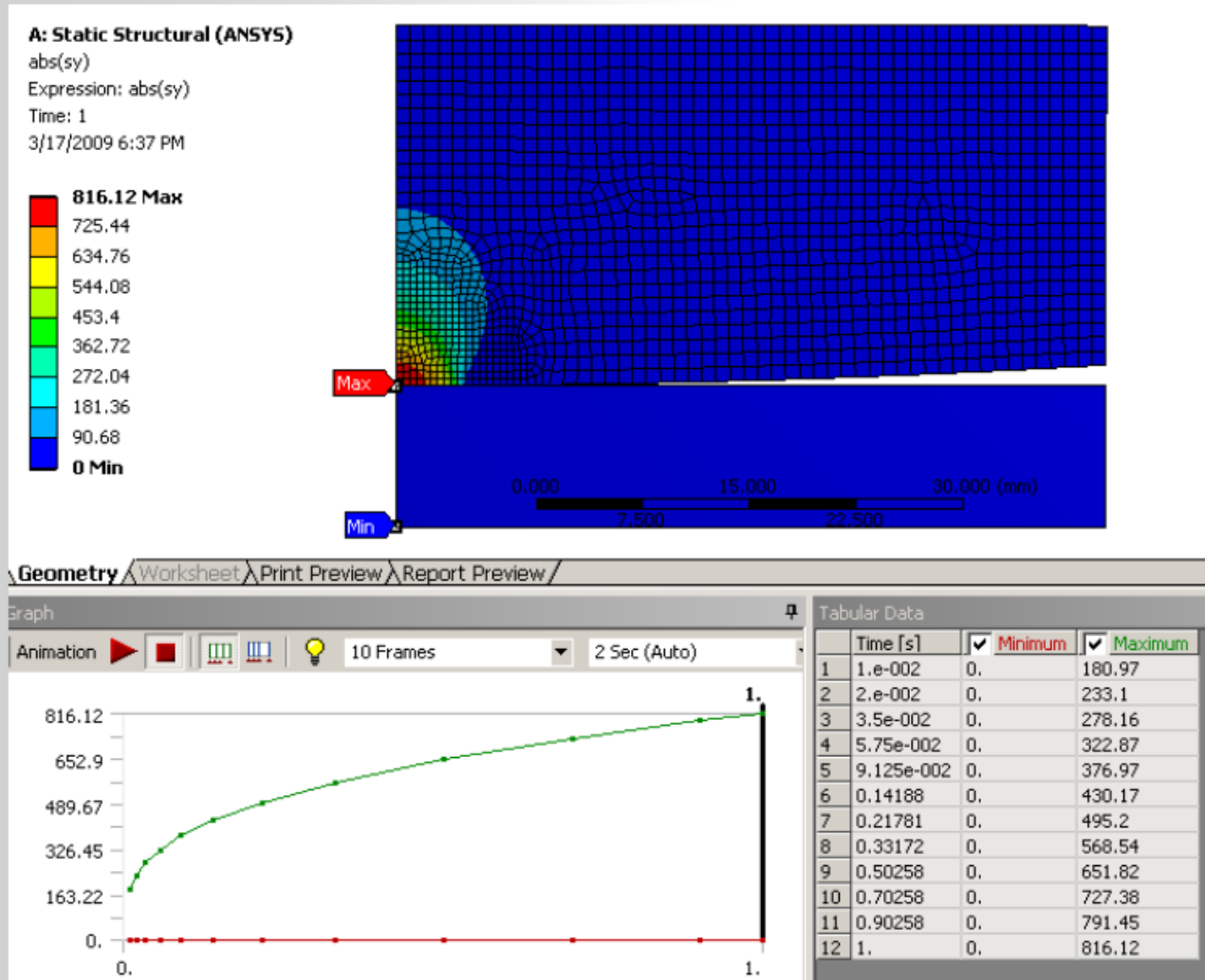
If you choose any of these stiffness update settings, the program will modify the stiffness (raise/lower/leave unchanged) based on the physics of the model (that is, the underlying element stress and penetration).

An advantage of choosing either of the program stiffness update settings is that stiffness is automatically determined that allows both convergence and minimal penetration.

G. Workshop – Contact Stiffness & Penetration

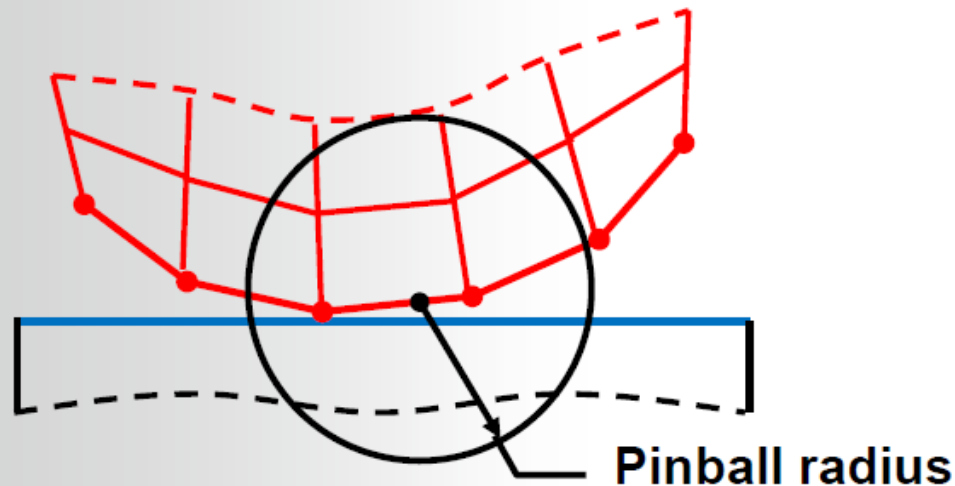
Please refer to your *Workshop Supplement* for instructions on:

Workshop 3A-Contact Stiffness Study



H. Pinball Region

- The Pinball Region is a contact element parameter that differentiates between far field open and near field open status. It can be thought of as a spherical boundary surrounding each *contact detection point*
 - If a detection point is close to the target surface within this sphere, *Mechanical* considers it to be in a “near” open condition and will record and track the gap more closely (i.e., when and whether contact is established). Gaps for detection point away from the target a distance larger than this sphere outside of this sphere will not be recorded or monitored.
 - If Bonded Behavior is specified within a gap smaller than the Pinball Radius, *Mechanical* will still treat that region as bonded, but ignore the gap.



... Pinball Region

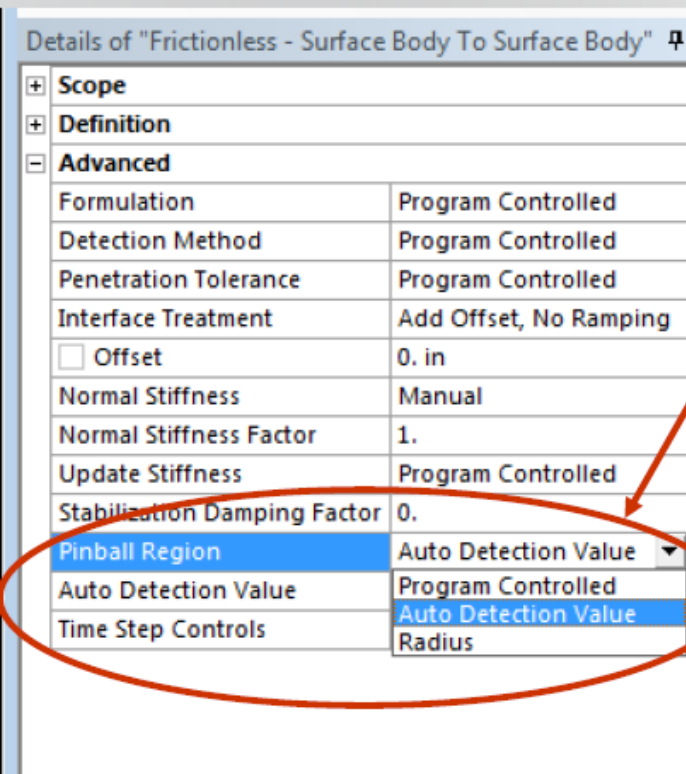
There are several uses for the Pinball Region:

- Provides computational efficiency in contact calculations, by differentiating “near” and “far” open contact when searching for which possible elements can contact each other in a given Contact Region.
- Determines the amount of allowable gap for bonded contact. If MPC Formulation is active, it also affects how many nodes will be included in the MPC equations.
- Determines the depth at which initial penetration will be resolved if present

For bonded and no separation [contact types](#), you must be careful in specifying a large pinball region. For these types of contact, any regions found within the pinball region will be considered to be in contact. For other types of contact, this is not as critical because additional calculations are performed to determine if the two bodies are truly in contact.

... Pinball Region

There are three options for controlling the size of the Pinball Region for each contact detection point.

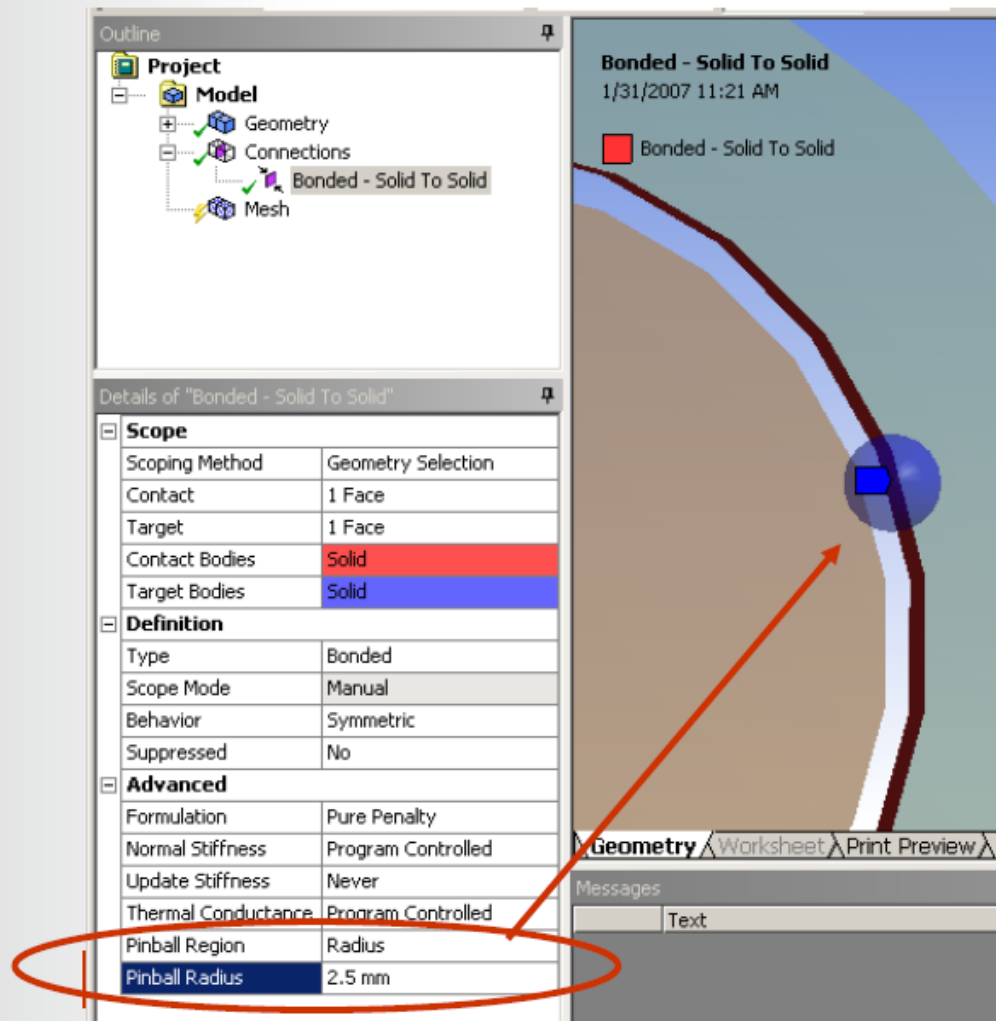


- **Program Controlled - (default)** The pinball region will be calculated by the program based on underlying element type and size.
- **Auto Detection Value -** The pinball region will be equal to the Tolerance Value as set on the Global Contact Settings.
 - Ensures that contact pairs created through the automatic contact detection have a Pinball Radius that envelops gap between target and contact.
 - Recommended option for cases where the automatic contact detection region is larger than the program controlled pinball value. In such cases, some contact pairs that were detected automatically may not be initially closed at start of solution.
- **Radius -** User manually specifies a value for the pinball region.

... Pinball Region

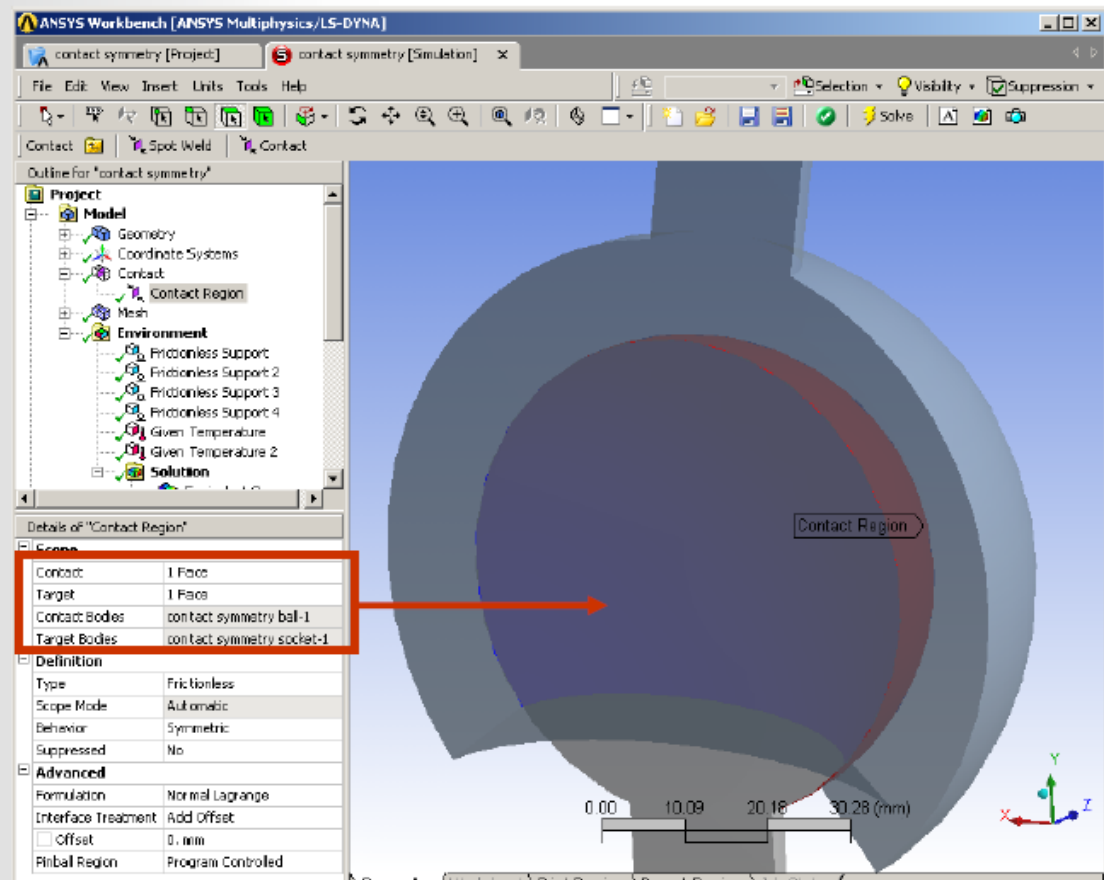
“Auto Detection Value” or a user defined Pinball “Radius” will appear as a sphere on the Contact Region for easy verification.

By specifying a Pinball Radius, one can visually confirm whether or not a gap will be ignored in Bonded Behavior. The Pinball Region can also be important in initial interference problems or large-deformation problems.



I. Symmetric/Asymmetric Behavior

- Internally, the designation of *Contact* and *Target* surfaces can be very important
 - In *Mechanical*, under each “Contact Region,” the *Contact* and *Target* surfaces are shown. The normals of the *Contact* surfaces are displayed in red while those of the *Target* surfaces are shown in blue.
 - The *Contact* and *Target* surfaces designate which two pairs of surfaces can come into contact with one another.



The concept of Symmetric vs Asymmetric *Behavior* only applies to penalty based methods



Definition	
Type	Bonded
Scope Mode	Automatic
Behavior	Program Controlled
Suppressed	Program Controlled
Advanced	
Formulation	Asymmetric
	Symmetric
	Auto Asymmetric

- *Asymmetric Behavior*

- Only the contact surfaces are constrained from penetrating the target surfaces.
- Internally, contact elements are meshed onto the red surface and corresponding target elements are meshed onto the blue surface, constituting one contact “pair”.

- *Symmetric Behavior.*

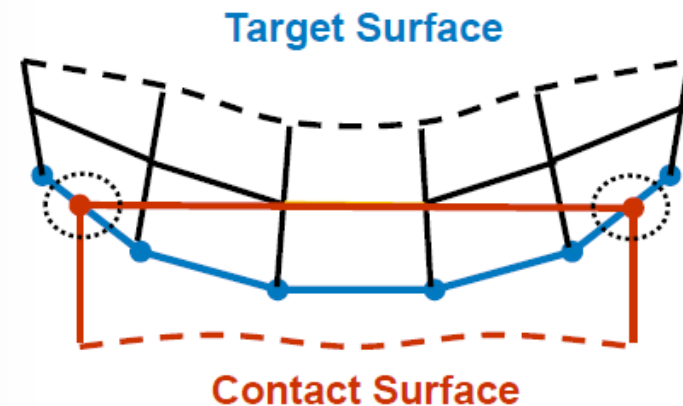
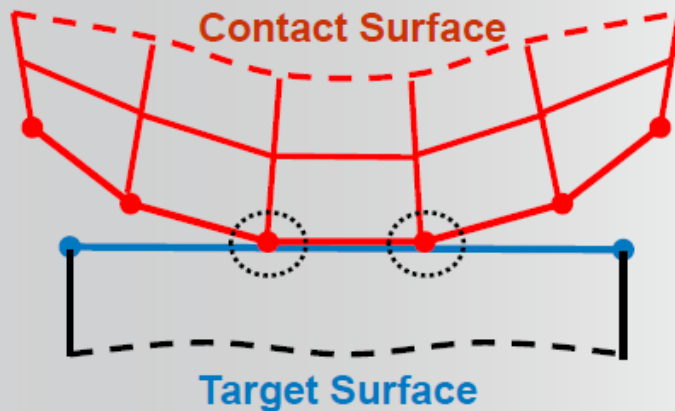
- The contact surfaces are constrained from penetrating the target surfaces *and* the target surfaces are constrained from penetrating the contact surfaces.
- Internally, the program uses two contact pairs with contact and target elements residing on both red and blue surfaces.

- *Auto-Asymmetric* (Default behavior with Program controlled option)

- The program evaluates the contact region and chooses which surface should be meshed with contact elements and which should be meshed with target elements.
- Internally, this may or may not result in one contact pair, but the contact elements may end up on the blue surface and target elements on the red surface.

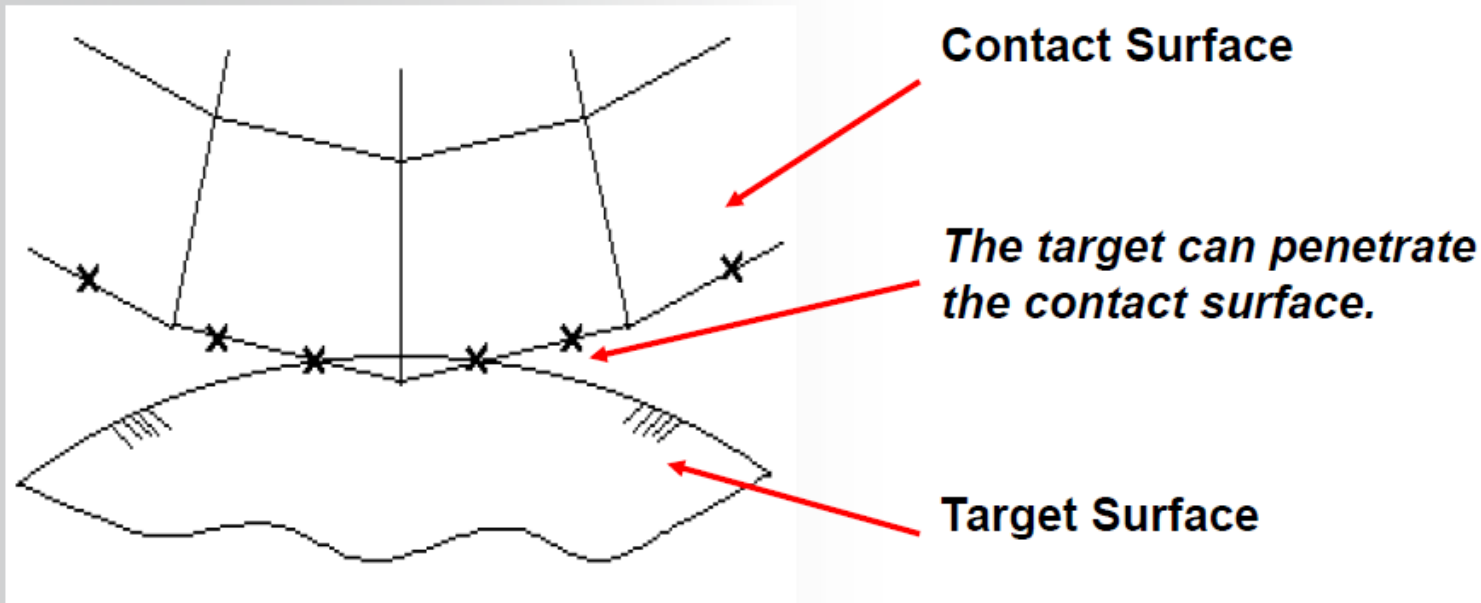
... Symmetric/Asymmetric Behavior

- For *Asymmetric Behavior*, the nodes of the *Contact* surface cannot penetrate the *Target* surface. This is a very important rule to remember. Consider the following:
 - On the left, the top red mesh is the mesh on the *Contact* side. The nodes cannot penetrate the *Target* surface, so contact is established correctly
 - On the right, the bottom red mesh is the *Contact* surface whereas the top is the *Target*. Because the nodes of the *Contact* cannot penetrate the *Target*, too much actual penetration occurs.



... Symmetric/Asymmetric Behavior

- Asymmetric Behavior (integration points only residing on one side), may allow some penetration at edges because of the location of contact detection points.
- The figure on the bottom illustrates this case:



... Symmetric/Asymmetric

- Only *Pure Penalty* and *Augmented Lagrange* formulations actually support Symmetric Behavior.
- *Normal Lagrange* and *MPC* require Asymmetric Behavior.
 - Because of the nature of the equations, Symmetric Behavior would be overconstraining the model mathematically, so Auto-Asymmetric Behavior is used even when Symmetric Behavior is selected.
- It is always good for the user to follow the general rules of thumb in selecting Contact and Target surfaces noted on the previous slide for any situation below where Asymmetric Behavior is used.

	Specified Option	Pure Penalty	Augmented Lagrange	Normal Lagrange	MPC
Behavior Internally Used	Symmetric Behavior	Symmetric	Symmetric	<i>Auto-Asymmetric</i>	<i>Auto-Asymmetric</i>
	Asymmetric Behavior	<i>Asymmetric</i>	<i>Asymmetric</i>	<i>Asymmetric</i>	<i>Asymmetric</i>
	Auto-Asymmetric Behavior	<i>Auto-Asymmetric</i>	<i>Auto-Asymmetric</i>	<i>Auto-Asymmetric</i>	<i>Auto-Asymmetric</i>
Reviewing Results	Symmetric Behavior	<i>Results on Both</i>	<i>Results on Both</i>	<i>Results on Either</i>	<i>Results on Either</i>
	Asymmetric Behavior	Results on Contact	Results on Contact	Results on Contact	Results on Contact
	Auto-Asymmetric Behavior	<i>Results on Either</i>	<i>Results on Either</i>	<i>Results on Either</i>	<i>Results on Either</i>
Notes	Symmetric Behavior	Easier to set up	Easier to set up	Let program designate	Let program designate
	Asymmetric Behavior	Efficiency and control	Efficiency and control	User has control	User has control
	Auto-Asymmetric Behavior	Let program designate	Let program designate	Let program designate	Let program designate

... Symmetric/Asymmetric

Symmetric Behavior:

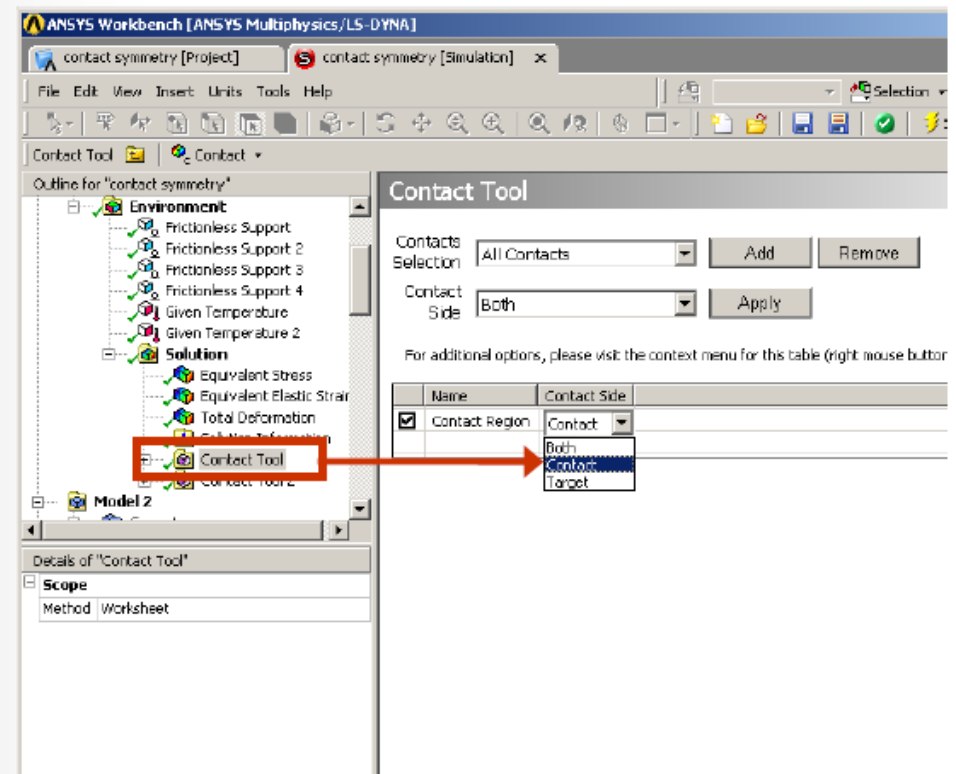
- Easier to set up
- More computationally expensive.
- Interpreting data such as actual contact pressure can be more difficult
 - Results are reported on both sets of surfaces

Asymmetric Behavior:

- Mechanical can automatically perform this designation (Auto-Asymmetric) or...
- User can designate the appropriate surface(s) for contact and target manually .
 - Selection of inappropriate Contact vs.Target may affect results.
- Reviewing results is easy and straightforward. All data is on the contact side.

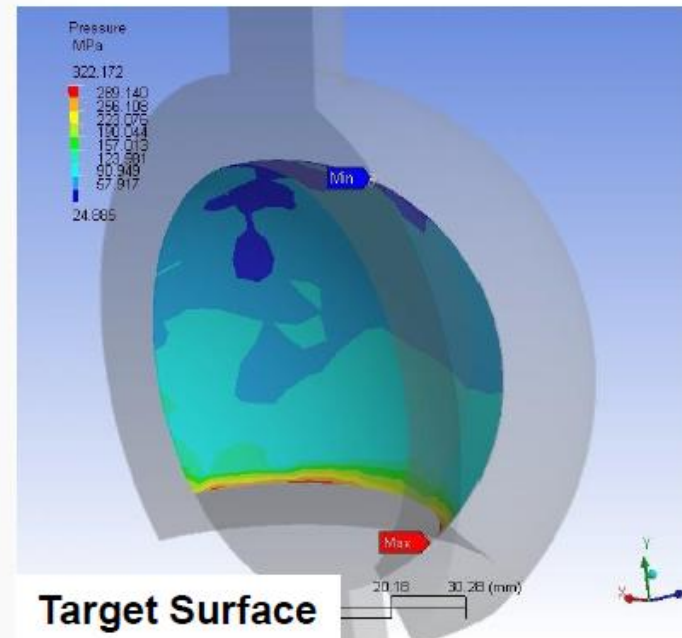
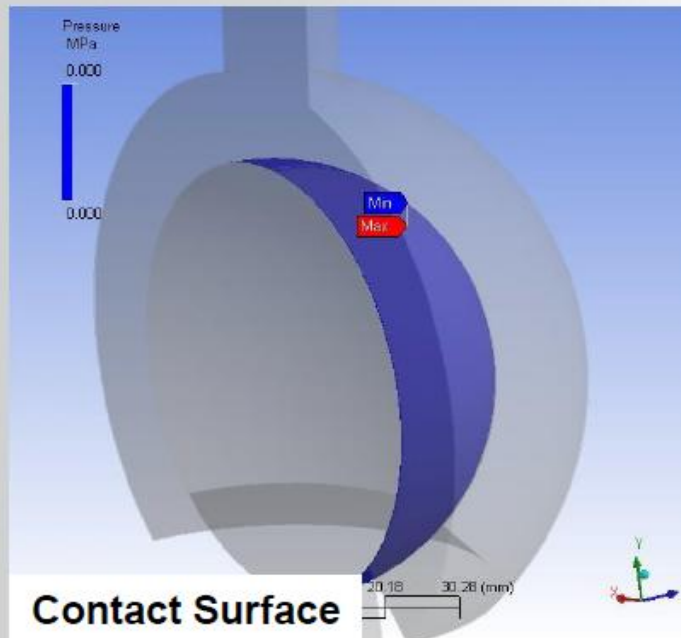
K. Contact results

- For Symmetric Behavior, results are reported for both Contact and Target surfaces.
- For any resulting Asymmetric Behavior, results are only available on Contact surfaces.
- When viewing the Contact Tool worksheet, the user may select Contact or Target surfaces to review results.
- For Auto-Asymmetric Behavior, the results may be reported on *either* the Contact or Target
- For Asymmetric Behavior, zero results are reported for Target



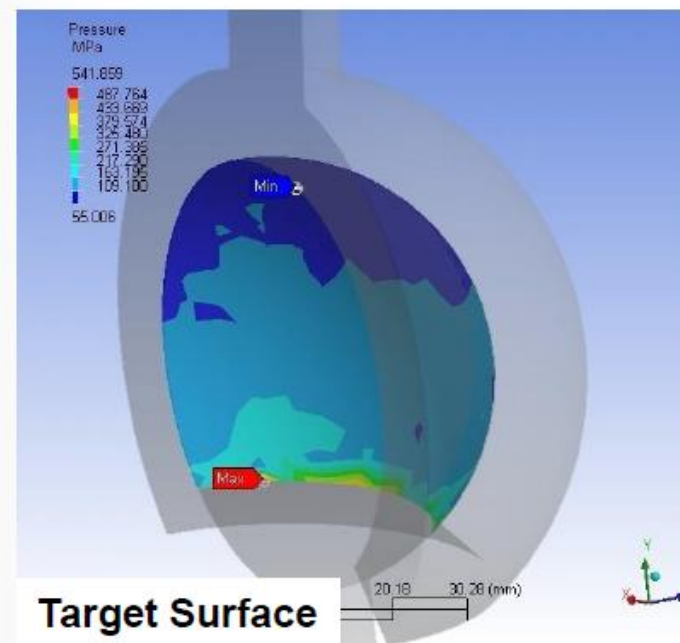
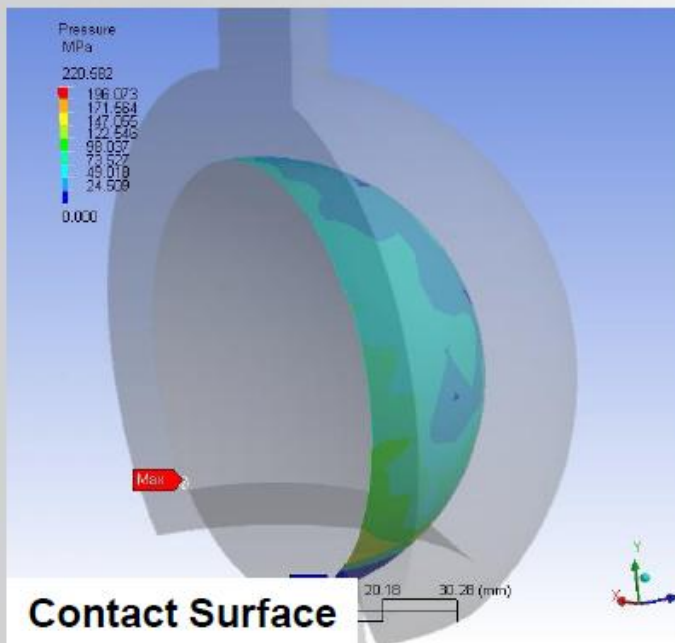
... Contact results

- For example, consider the case below of Normal Lagrange Formulation with Symmetric Behavior specified.
 - This results in auto-asymmetric behavior. Since it is automatic, WB-Mechanical may reverse the Contact and Target specification.
- When reviewing Contact Tool results, one can see that the Contact side reports no (zero) results while the Target side reports true Contact Pressure.



... Contact results

- In another situation, Augmented Lagrange Formulation with Symmetric Behavior is used
 - This results in true symmetric behavior, so both set of surfaces are constrained from penetrating each other
- However, results are reported on *both* Contact and Target surfaces. This means that the “true” contact pressure is an average of both results.



L. Workshop – Symmetric vs Asymmetric

Please refer to your *Workshop Supplement* for instructions on:

W3B Symmetric vs Asymmetric

